

DL report: 1_pytorch_intro

Hypotheses per parameter

Epochs

H_0 : The resulting accuracy will be higher when the data is trained with more epochs.

H_1 : The resulting accuracy will be lower or equal when the data is trained with more epochs.

An epoch is a complete pass through the all the training data. The more often a model in training is able to pass through the data, the more it will learn based on the data and its own (mis)calculations.

Result: Tests have been performed with different epochs (3, 5, 10, 15) and the resulting accuracy at the last epoch is overall a higher result. This does depend on the unit values. When the first and second unit values are equal to one another, the accuracy drops slightly between epoch 10 and 15, as seen in the table below.

		Epochs			
		3	5	10	15
Unit values	256, 256	0.7881	0.7959	0.8272	0.8266
	256, 128	0.7825	0.8006	0.8381	0.8506
	256, 64	0.7700	0.8063	0.8206	0.8369
	128, 256	0.7812	0.8131	0.8347	0.8375
	128, 128	0.7866	0.7981	0.8263	0.8244
	128, 64	0.7709	0.7875	0.8237	0.8456
	64, 256	0.7269	0.8022	0.8119	0.8384
	64, 128	0.7388	0.7878	0.8216	0.8159
	64, 64	0.7509	0.7969	0.8287	0.8247

Unit values

H_0 : The resulting accuracy will be higher when the data is trained with a higher number of units in the layers.

H_1 : The resulting accuracy will be lower or equal when the data is trained with a higher number of units in the layers.

The higher the number of units in the layers, the more complex the model will become. If the model is more complex it is able to learn/ recognize more complex patterns in the data.

Result: Tests have been performed with unit values [64, 32, 16], [256, 128, 64] and [1024, 512, 256] and the resulting accuracy is overall higher when the number of units in layers are higher as seen in the table below.

		Unit values		
		[256, 128, 64]	[64, 32, 16]	[1024, 512, 256]
Loop	1	0.7881	0.7534	0.8109
	2	0.7825	0.7559	0.7925
	3	0.7700	0.6637	0.8131
	4	0.7812	0.7453	0.7647

	5	0.7866	0.7069	0.7953
	6	0.7709	0.7084	0.7891
	7	0.7269	0.7253	0.7872
	8	0.7388	0.6787	0.8025
	9	0.7509	0.6534	0.7581

Batchsize

H_0 : The resulting accuracy will be higher when the batchsize is higher.

H_1 : The resulting accuracy will be lower or equal when the batchsize is higher.

A batch is a subset of the complete training set. If the size of the batch is larger, then more training data is available for the model to train on and learn from.

Result: Tests have been performed with batch sizes 32, 64 and 128. The resulting accuracy is mostly higher when the batch size is higher.

Correlation

While evaluating test results a correlation between accuracy and unit values has been discovered. When the second unit is higher or equal in size compared to the first unit, there is a higher chance that the accuracy of the model will drop when increasing other parameters.